A No-Go Diagnostic for DiT-Only Structured Pruning of Wan Image-to-Video Generation

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Abstract

Deploying image-to-video diffusion transformers requires compression methods that reduce repeated denoiser cost without breaking subject fidelity, temporal behavior, or image quality under the same benchmark contract used for the uncompressed model. Existing diffusion-efficiency work does not establish whether Wan2.1-Fun-V1.1-1.3B-InP can meet this standard when the intervention is restricted to structured DiT pruning, unlike sampling acceleration, token skipping, or dynamic routing, and when the VAE, text encoder, image encoder, tokenizer, scheduler, and other non-DiT modules must remain unchanged. We evaluate a structured 24-layer, 1344-hidden, 7168-FFN, 12-head DiT candidate that removes source layers 12–17, applies width reduction, keeps the Wan I2V inference backend otherwise fixed, and scores the step-100 recovery checkpoint with the official non-camera VBench-I2V evaluator over 355 matched rows. The candidate reaches 920,852,288 DiT parameters, below the 1.0B target, but the camera-excluded aggregate drops from 0.866734 for the unpruned baseline to 0.702540 for the recovered candidate, a relative degradation of 18.944% rather than the required 3–5%. This measured no-go result shows that parameter-count success is insufficient for Wan I2V pruning, and it identifies the next evidence needed for a positive claim: less aggressive sub-1B candidates, longer recovery, ablations separating depth and width, and resolution of evaluator variant warnings.

Introduction

Video diffusion models increasingly rely on transformer denoisers, inheriting the scaling behavior of diffusion transformers while adding temporal and conditioning demands (Ho, Jain, and Abbeel 2020; Rombach et al. 2021; Blattmann et al. 2023; Peebles and Xie 2023; Wan Team 2025). For image-to-video deployment, the denoiser is the natural compression target: it dominates repeated denoising computation, while peripheral modules such as tokenizers, encoders, schedulers, and decoders are shared across many generation settings. The project target in this paper is consequently narrow and measurable: prune only the DiT denoiser of Wan2.1-Fun-V1.1-1.3B-InP, keep non-DiT modules unpruned, reach at most 1.0B DiT parameters if possible, and keep official non-camera VBench-I2V degradation within 3–5% relative to the unpruned baseline.

Recent diffusion-efficiency work shows that the family of possible interventions is broad. PFDiff accelerates sampling without changing the denoiser weights (Wang et al. 2024), SparseDiT and CoReDiT reduce token computation inside diffusion transformers (Chang et al. 2024; Li et al. 2026), and UniCP combines caching with pruning-style execution savings for video generation (Sun et al. 2025). Those approaches are relevant to deployment cost, but they do not by themselves satisfy this project's static-model target: the delivered model must be a smaller DiT, not only a faster sampler or a dynamically skipped computation graph. This gap motivates the structured DiT-only test reported here, where the previewed result is deliberately negative: 920,852,288 DiT parameters but 18.944% official non-camera VBench-I2V degradation.

The verified outcome is negative but useful. The current candidate reaches the parameter side of the target, with 920,852,288 measured DiT parameters. It does not reach the quality side: official non-camera VBench-I2V aggregate degradation is 18.944%, far above the intended 3–5% budget. This paper therefore records a diagnostic result rather than a success claim. Its contributions are: (1) a reproducible DiT-only structured pruning candidate for Wan I2V; (2) an official non-camera VBench-I2V matched evaluation report; (3) a disciplined separation between supported parameter claims and unsupported quality-success or causal claims; and (4) a concrete list of missing ablations and protocol checks required before the next pruning candidate can support a stronger claim.

The negative framing is important for two reasons. First, a sub-1B DiT count alone is not a deployment result for image-to-video generation: the value of compression depends on retaining subject binding, background faithfulness, temporal behavior, and image quality under the same evaluation contract used for the baseline. Second, the practical search space contains many plausible confounds. A failed candidate could be too narrow, too shallow, insufficiently recovered, mismatched to the recovery distribution, or evaluated under a protocol that needs additional variant handling. Reporting those distinctions explicitly is more useful than collapsing them into a broad statement that structured pruning does or does not work for Wan. The paper therefore treats the measured run as a bounded diagnostic: it is strong enough to reject this candidate as a target-satisfying result, but not strong enough to reject all near-1B candidates or all recovery schedules.

Related Work

Diffusion and video foundations. Denoising diffusion models formulate generation as iterative denoising from a noisy latent or image trajectory (Ho, Jain, and Abbeel 2020). Latent diffusion reduces image-space cost by denoising in a compact latent space (Rombach et al. 2021), and video latent diffusion extends this setting to temporal outputs (Blattmann et al. 2023). Diffusion transformers replace convolutional denoisers with transformer backbones and expose scaling behavior through attention and feed-forward width (Peebles and Xie 2023). Wan provides an open video-generation family that includes the model lineage targeted here (Wan Team 2025). These foundations motivate DiT-specific compression, but they do not by themselves establish that an aggressively pruned Wan I2V denoiser will retain benchmark quality.

The distinction between image and image-to-video compression matters here. A pruned image generator can lose fine texture or prompt alignment while still producing a single plausible frame. In image-to-video generation, the denoiser must preserve the conditioning image, maintain subject and background identity over time, and produce motion without introducing temporal defects. A DiT pruning configuration that looks reasonable from parameter accounting alone can therefore fail through several benchmark channels at once. This is why the paper uses VBench-I2V dimensions rather than a single visual score or a generic sampling-speed proxy.

Structured diffusion-transformer compression. Structured pruning is attractive because it can produce a deployably smaller network rather than a sparse mask. PPCL-style contiguous layer pruning and distillation are directly relevant to the depth component of our candidate (Ma et al. 2025). Post-training DiT pruning addresses compression after pretraining (Hu et al. 2026), while PARE studies pruning and adaptive routing for video generation rather than a fixed Wan I2V parameter target (Wang et al. 2026). TinyFusion and HierarchicalPrune study shallow or position-aware diffusion-model compression (Fang et al. 2024; Kwon et al. 2026). These papers motivate the design space but leave a narrower unanswered question: whether a Wan I2V DiT-only candidate that combines depth and width pruning can meet an official non-camera VBench-I2V retention budget while keeping every non-DiT module unpruned.

The candidate in this paper combines two structured edits that are often studied separately: removing entire transformer blocks and narrowing channel dimensions. That combination is attractive because it lowers both repeated block cost and per-block matrix dimensions, but it also makes attribution difficult. If quality falls, the evidence must distinguish among lost depth, reduced representation width, reduced feed-forward capacity, changed attention head dimension, recovery insufficiency, and evaluation-protocol effects. The current run does not contain those controls. The related work therefore serves as design context, not as evidence that the present architecture should have worked.

Video-generation evaluation. Video-generation evaluation is multi-dimensional: subject faithfulness, background faithfulness, temporal stability, motion, aesthetics, and imaging quality can move in different directions. VBench provides a benchmark suite designed to decompose video quality into such dimensions (Huang et al. 2024a), and VBench++ extends the evaluation setting further (Huang et al. 2024b). This paper uses the official VBench-I2V dimensions required by the project, excluding camera/control camera-motion metrics and retaining the other I2V dimensions. The evaluation is not a human study and is not a claim of perceptual equivalence; it is a benchmark-grounded no-go test for a specific candidate.

The exclusion of camera/control camera-motion metrics is not a post-hoc filter. It is part of the project contract because the target is image-to-video quality under the official non-camera I2V dimensions. Including camera-control metrics would change the question by rewarding behavior outside the intended evaluation surface. Conversely, reporting only the aggregate would hide the diagnostic structure needed for a pruning paper. The table and figures therefore preserve the full non-camera dimension breakdown while keeping camera metrics absent.

Negative results and reproducible compression claims. Compression papers often emphasize the best retained-quality point, but a negative point can be equally informative when it is tied to a clear target and a replayable evidence chain. In this study, the claim boundary is deliberately narrow: the current result rejects one evaluated candidate under one official non-camera VBench-I2V contract. The paper does not claim that the dynamic-degree gain compensates for semantic and visual losses, does not treat within-run bootstrap diagnostics as multi-seed significance, and does not attribute the failure to one pruning component without ablations. This claim discipline is part of the method because the research question is not only whether a smaller network can be built, but whether the resulting evidence can support a deployable pruning claim.

Method

The evaluated candidate applies structured pruning only to the DiT denoiser. It keeps 24 of the 30 source DiT layers and removes one contiguous middle interval, source layers 12 through 17. Width pruning then reduces the hidden dimension from 1536 to 1344, the feed-forward dimension from 8960 to 7168, and the attention head dimension from 128 to 112 while retaining 12 heads. We denote this evaluated architecture as the 24-layer, 1344-hidden, 7168-FFN, 12-head candidate. The measured DiT parameter count is 920,852,288.

Depth pruning. The depth edit is a contiguous middle-block removal rather than a scattered layer mask. The retained source-to-target mapping preserves source layers 0–11, skips source layers 12–17, and maps source layers 18–29 to target layers 12–23. This choice follows the intuition that early and late denoiser blocks may carry input-conditioning and output-refinement roles that are risky to remove blindly.

The paper does not claim this interval is optimal; it is the specific interval used by the evaluated candidate.

Width pruning. The width edit reduces the transformer state and feed-forward expansion while keeping the number of attention heads fixed. Holding 12 heads while reducing the hidden dimension changes the per-head dimension from 128 to 112. The feed forward dimension is reduced from 8960 to 7168. This makes the candidate a combined depth-plus-width point rather than an isolated test of either axis. Because no width-only or depth-only evaluation exists, any interpretation of which edit damaged which metric remains a hypothesis rather than a supported claim.

Parameter accounting. Parameter counts are computed on the instantiated baseline and evaluated candidate rather than extrapolated from nominal layer dimensions alone. The unpruned DiT contains 1,564,428,608 parameters. The recovered candidate contains 920,852,288 DiT parameters, and this is the candidate scored in the matched VBench-I2V comparison. The parameter row in Table 2 therefore reports a measured model property rather than a paper-only estimate.

Non-DiT modules are intentionally outside the pruning surface. The VAE, text encoder, image encoder, tokenizer, scheduler, and other non-DiT references remain tied to the unpruned base model. This matters for interpretation: any measured quality loss is attributed to the smaller denoiser candidate and its recovery procedure, not to a smaller VAE or changed conditioning stack.

The unchanged-module policy is also a fairness control. If the VAE or encoders were pruned at the same time, a large loss in subject or image quality could not be localized to the denoiser candidate. By fixing the peripheral stack, the experiment asks a cleaner question: whether this smaller DiT, plugged back into the same Wan I2V system, retains enough benchmark quality. This does not mean the denoiser is the only possible deployment bottleneck; it means the current paper isolates the only pruning surface allowed by the project contract.

The candidate includes a recovery pilot before benchmark scoring. The recovery checkpoint evaluated here is the saved step-100 checkpoint. Because no longer-recovery evaluation is available, the paper does not claim that recovery has converged or that additional distillation would be ineffective. Similarly, because there is no depth-only, width-only, or less-aggressive sub-1B ablation matrix, the paper does not claim that a particular pruning component caused the observed failure.

This recovery status is a central limitation rather than a footnote. A short recovery pilot can reveal whether the candidate is immediately viable, but it cannot settle whether the architecture would recover under a longer schedule, different data curriculum, or distillation target. The paper therefore uses "recovered pruned candidate" only to identify the evaluated checkpoint; it does not use the phrase to imply full convergence or recovery sufficiency.

Why this candidate is still a valid test. Although the candidate is not sufficient for a success claim, it is still a valid diagnostic. The architecture satisfies the structural scope: only the DiT is pruned, the non-DiT modules are held fixed, and the measured DiT count is below the target. The evaluation satisfies the matched-run scope: baseline and pruned videos are paired by prompt/image condition and generation settings. The result therefore answers a concrete question: whether this particular 24-layer, 1344-hidden, 7168-FFN candidate with step-100 recovery is inside the target quality band. The answer is no. That negative answer narrows the search space for the next candidate without pretending to solve the broader compression problem.

Figure 1 summarizes the tested pipeline and the resulting no-go verdict. Critical numerical claims remain in LaTeX text and tables rather than relying only on bitmap labels.

Experimental Setup

The paper-facing system identifier is the Wan2.1-Fun-V1.1-1.3B-InP Wan I2V inference pipeline evaluated by the official VBench-I2V harness. The benchmark compares the unpruned baseline against the recovered pruned candidate using matched prompts, images, generation settings, and seed. The official non-camera I2V evaluation contains 355 matched task rows, no camera rows, and no baseline/pruned generation-setting mismatches. It reports one baseline score and one recovered-candidate score for each of the nine required non-camera dimensions.

Matched comparison contract. Each paper-facing task row contains the prompt/image condition and paired baseline and pruned videos for the same comparison unit. The evaluated split has 355 expected rows, 355 task rows, zero missing images, zero duplicated row identifiers, and zero generation-setting mismatches. This matters because a large aggregate loss would be difficult to interpret if the baseline and pruned conditions used different prompts, resolutions, frame counts, denoising steps, or guidance values. The current comparison is still single-seed, but it is matched within that seed.

The dimensions are *i2v subject*, *i2v background*, *subject consistency*, *background consistency*, *temporal flickering*, *motion smoothness*, *dynamic degree*, *aesthetic quality*, and *imaging quality*. Camera and control camera-motion dimensions are excluded by design. The aggregate reported here is the camera-excluded weighted aggregate computed from those official dimensions.

The nine retained dimensions cover complementary failure modes. The two I2V faithfulness dimensions score how well generated videos preserve subject and background information from the conditioning image. Subject and background consistency track stability of those elements over time. Temporal flickering and motion smoothness capture temporal artifacts. Dynamic degree measures whether motion is present, while aesthetic and imaging quality capture visual appeal and low-level image quality. This decomposition is why the paper reports per-dimension rows rather than only the aggregate: a candidate can improve one dimension, such as dynamic degree, while still failing the overall quality budget through semantic and visual regressions.

The run is deliberately treated as single-seed evidence.

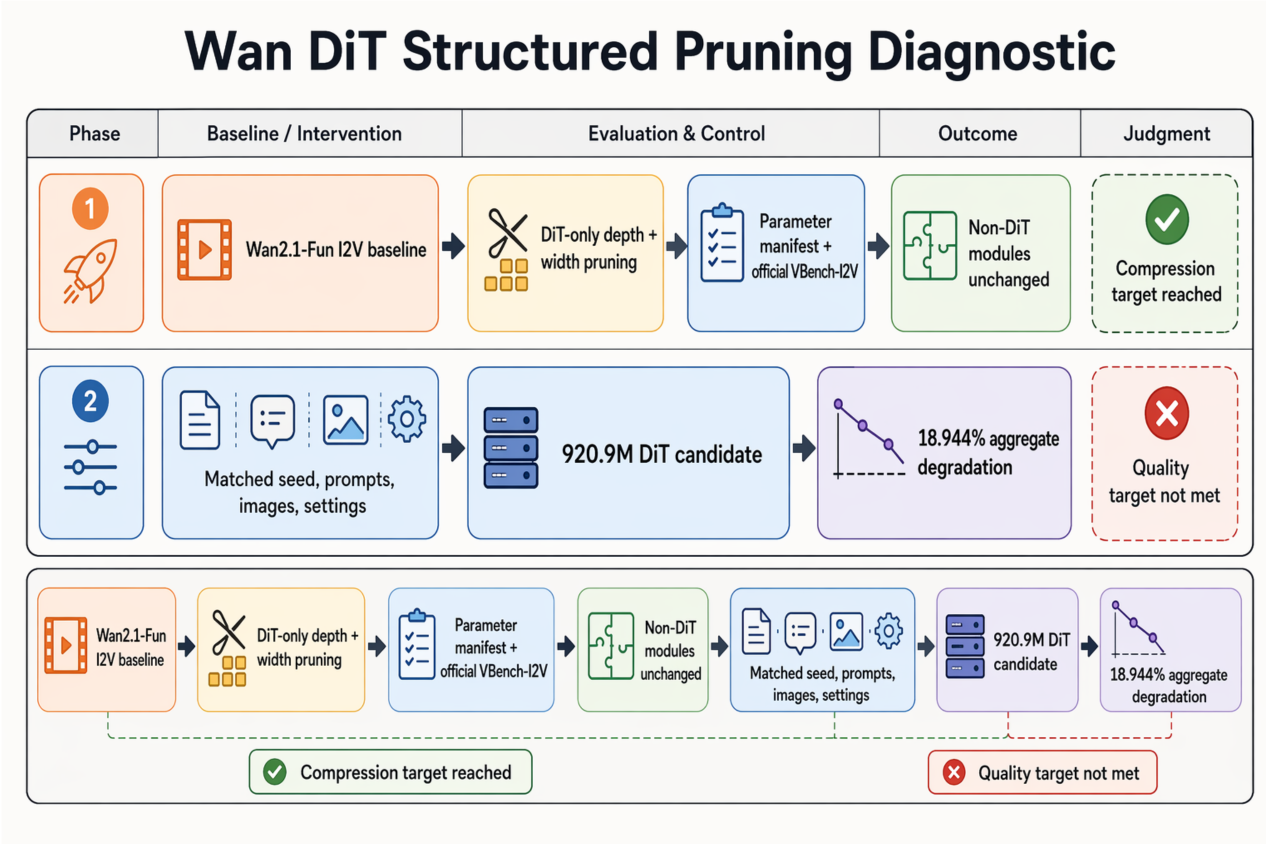

Figure 1: Structured-pruning diagnostic overview. The evaluated setup preserves the non-DiT stack, applies DiT-only depth and width pruning, verifies the 920.9M-parameter candidate with matched official non-camera VBench-I2V scoring, and records the resulting 18.944% aggregate degradation as a no-go rather than a positive pruning claim.

Both baseline and pruned generations use seed 43, 81 frames, 50 denoising steps, 832 by 480 resolution, and guidance 6.0. These controls make the candidate comparison interpretable, but they do not provide multi-seed robustness. The official evaluator log also reports missing-video warnings for expected suffix variants. The generated video sets each contain 355 videos and zero missing generated videos, so this warning is a protocol-confidence gap rather than a reason to silently discard the score.

Evaluator warning policy. The warning policy is conservative. The score is retained because the official scoring run completed, produced finite non-constant values, and matched the generated videos that were actually produced. At the same time, the warning is not hidden: evaluator stdout reports missing expected suffix variants for '-1' through '-4', while the generation outputs contain one video per prompt. Until the one-video contract is justified for this I2V path or the expected variants are regenerated, the score is best read as official but protocol-warned evidence.

Aggregate computation and target test. The aggregate is the camera-excluded weighted aggregate over the retained official dimensions. Eight dimensions have weight 1.0 and dynamic degree has weight 0.5, giving an aggregate weight of 8.5. The target test is relative to the unpruned baseline, not an absolute recovered score threshold. Under that test, the baseline aggregate is 0.866734090105 and the recovered aggregate is 0.702539977713, giving relative degradation 0.189440007340. Because this is far above 0.05, the candidate fails even under the high end of the allowed loss budget.

Evidence Chain

The evidence chain has four layers. The first layer is the project contract: structured pruning only, DiT-only pruning surface, non-DiT modules unchanged, sub-1B DiT target, and official non-camera VBench-I2V evaluation. The second layer is the parameter count: the base DiT is measured at 1,564,428,608 parameters and the recovered pruned DiT is measured at 920,852,288 parameters. The third layer is the matched evaluation split: 355 task rows, zero camera rows, zero missing images, zero duplicated row identifiers, and zero baseline/pruned setting mismatches. The fourth layer is official scoring: finite, varying scores for the baseline and recovered conditions across the nine retained dimensions.

This layering is what allows the manuscript to make a precise negative claim. If the parameter count were missing, the paper could not say the candidate actually meets the sub-1B

Benchmark/source	Model/backend	Task count/split	Role/method	Budget/decoding policy	Metric	Numerical takeaway
Official non-camera VBench-I2V evaluator	Wan2.1-Fun-V1.1-1.3B-InP Wan I2V pipeline	355 matched rows; 0 camera rows	Unpruned baseline	Seed 43; 81 frames; 50 steps; 832×480; guidance 6.0	Camera-excluded aggregate	0.866734 reference
Official non-camera VBench-I2V evaluator	Same Wan I2V pipeline with 24L/1344H/7168F DiT	Same rows, prompts, images, and settings	DiT-only structured pruning plus step-100 recovery; non-DiT unchanged	Same seed, frames, steps, resolution, and guidance	DiT parameters and aggregate	920,852,288 DiT params; 0.702540 aggregate; 18.944% no-go

Table 1: Evaluation contract and numerical outcome. This matrix states the benchmark harness, evaluated model/backend, task split, baseline and method roles, generation budget, metrics, and key scores for the comparison.

DiT side of the target. If the non-DiT policy were unclear, the quality loss could be attributed to changed encoders or decoders. If the matched evaluation split contained camera leakage or setting mismatches, the aggregate would not answer the intended I2V question. If the official scoring result were absent, the quality loss would be a local proxy rather than the requested benchmark result. All four layers are present, but the quality layer rejects the candidate.

The same chain also prevents overclaiming. The result is not a statement about all structured pruning methods, all Wan checkpoints, all recovery schedules, or all possible sub-1B architectures. It is a statement about the evaluated candidate under the verified evaluation contract. This is the appropriate claim size for a negative result: narrow enough to be true, but detailed enough to guide the next experiment.

Results

Table 2 gives the headline result and all nine official non-camera VBench-I2V dimensions. The unpruned baseline has a camera-excluded aggregate score of 0.866734, while the recovered pruned candidate scores 0.702540. The relative degradation is 0.189440, or 18.944%. This is outside the allowed 3–5% budget, so the current candidate is a measured no-go.

The parameter and quality rows in Table 2 show the central tension. On the compression side, the candidate is below the 1.0B DiT threshold by 79,147,712 parameters. On the quality side, the aggregate loss is roughly four to six times larger than the allowed 3–5% band, depending on whether the low or high end of the target is used. The result is therefore not a marginal miss that could be rounded into compliance. It is a clear rejection of this candidate under the stated target.

The per-dimension rows in Table 2 show that the failure is not uniform. Temporal flickering and motion smoothness have small relative degradation in this run, and dynamic degree increases. In contrast, aesthetic quality degrades by 53.464%, imaging quality by 40.822%, i2v subject by 38.362%, subject consistency by 33.440%, and i2v background by 26.687%. These dimensions are the priority targets for the next candidate or recovery schedule.

Interpreting the retained temporal metrics. The small temporal flickering and motion-smoothness changes prevent

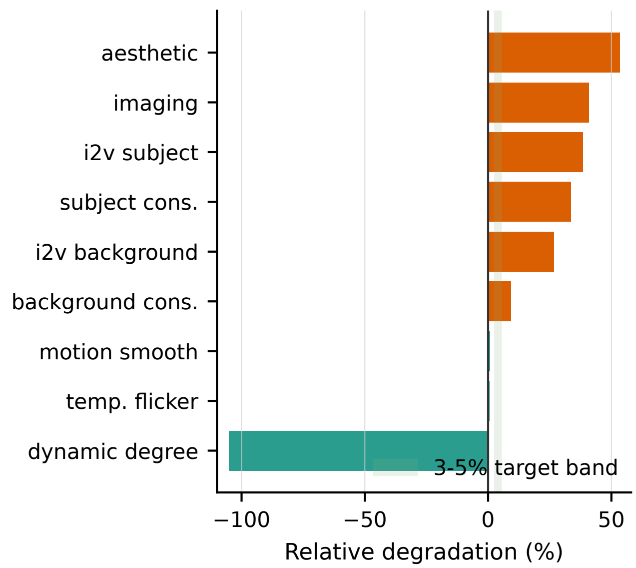

Figure 2: Relative degradation by VBench-I2V dimension shows that the no-go aggregate is driven mainly by aesthetic quality, imaging quality, i2v subject, subject consistency, and i2v background.

an over-simple interpretation that the pruned model merely becomes temporally unstable. In this run, the larger damage appears in semantic alignment and image quality. That distinction is useful for the next experiment: a less aggressive candidate or longer recovery should be judged not only by whether the aggregate improves, but by whether subject, background, aesthetic, and imaging scores recover without sacrificing the temporal metrics that were nearly retained.

Interpreting dynamic degree. Dynamic degree improves from 0.487805 to 1.000000 in this single run. The paper does not treat that improvement as evidence that the pruned model is better for I2V generation overall. Dynamic degree has weight 0.5 in the reported aggregate, whereas the large semantic and visual-quality losses occur across multiple weight-1 dimensions. The aggregate no-go verdict is therefore robust to the direction of the dynamic-degree row.

Metric or dimension	Benchmark/source	Rows	System and budget	Wt.	Baseline	Pruned	Change	Verdict
DiT parameters	Parameter counts	–	Wan2.1-Fun DiT; non-DiT unchanged	–	1,564,428,608	**920,852,288**	-41.137%	Parameter target met
Camera-excluded aggregate	Official VBench-I2V non-camera	355	Wan baseline vs. 24L/1344H/7168F DiT; seed 43, 81f, 50 steps, 832×480, guidance 6.0	8.5	**0.866734**	0.702540	18.944%	Quality no-go
i2v subject	Same official split	355	Same matched backend and budget	1.0	**0.946747**	0.583561	38.362%	Regression
i2v background	Same official split	355	Same matched backend and budget	1.0	**0.991421**	0.726843	26.687%	Regression
subject consistency	Same official split	355	Same matched backend and budget	1.0	**0.950742**	0.632812	33.440%	Regression
background consistency	Same official split	355	Same matched backend and budget	1.0	**0.971705**	0.882979	9.131%	Regression
temporal flickering	Same official split	355	Same matched backend and budget	1.0	**0.977198**	0.971633	0.570%	Near retained
motion smoothness	Same official split	355	Same matched backend and budget	1.0	**0.985352**	0.978841	0.661%	Near retained
dynamic degree	Same official split	355	Same matched backend and budget	0.5	0.487805	**1.000000**	-105.000%	Improved in this run
aesthetic quality	Same official split	355	Same matched backend and budget	1.0	**0.589287**	0.274232	53.464%	Largest loss
imaging quality	Same official split	355	Same matched backend and budget	1.0	**0.710885**	0.420689	40.822%	Large loss

Table 2: Main setup and score matrix for the evaluated candidate. Camera and control camera-motion metrics are absent by design. The table exposes the benchmark source, row count, evaluated Wan backend, generation budget, baseline and pruned/recovered scores for all nine non-camera VBench-I2V dimensions, the aggregate quality verdict, and the DiT parameter verdict. Bold marks the better score for each official dimension and the parameter value that satisfies the DiT budget; the aggregate row shows that the candidate is still a no-go because quality degradation is 18.944%, outside the 3–5% target.

Dimension ranking as engineering signal. The largest relative loss is aesthetic quality, followed by imaging quality, i2v subject, subject consistency, and i2v background. This ranking suggests that the evaluated denoiser reduction harms visual fidelity and conditioning more than it harms temporal smoothness. That observation matters for future recovery design. A recovery run that optimizes only temporal stability could leave the dominant aggregate losses untouched. A next candidate should instead track the five largest-loss dimensions as first-class acceptance criteria while also checking that the near-retained temporal dimensions do not regress.

Why the quality miss is decisive. The target band is 3–5% relative degradation. The measured aggregate loss is 18.944%. Even if one considers the lower-weight dynamic-degree improvement beneficial, the aggregate already incorporates that improvement and still fails. Even if the small temporal losses are viewed as encouraging, they do not compensate for the larger semantic and visual-quality losses. The candidate is therefore not a borderline result that requires subjective interpretation to classify. It is below the parameter limit and outside the quality limit.

Analysis / Ablation

The current evidence supports a negative result and a diagnostic ranking of failure dimensions. It does not support a causal ablation claim. There is no completed matrix comparing depth-only pruning, width-only pruning, combined pruning without recovery, longer recovery, or a less-aggressive candidate nearer to the 1.0B boundary. Consequently, this paper avoids statements such as “width pruning caused the aesthetic loss” or “contiguous layer pruning is insufficient for Wan I2V.” Those statements require experiments not yet present in the repository.

The most plausible reading is that the current candidate is too aggressive for the available recovery budget or that its recovery distribution is not aligned tightly enough with final I2V evaluation. The evidence does not distinguish between those explanations. The candidate removes six middle layers and reduces both hidden and FFN dimensions, so the failure may come from depth, width, their interaction, recovery length, or conditioning mismatch. A less-aggressive candidate below 1.0B parameters is a particularly important missing control because the evaluated model is 920.9M parameters, leaving some room before the target boundary.

Missing ablation matrix. A minimum follow-up matrix would include at least four classes of controls. A depth-

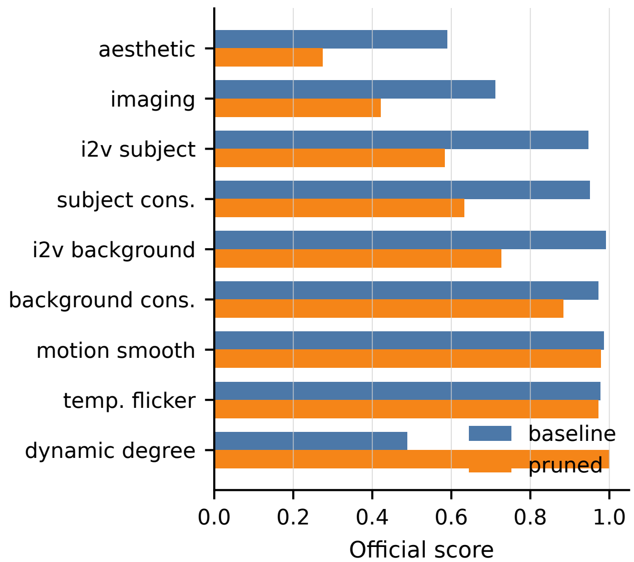

Figure 3: Baseline and recovered scores by official non-camera VBench-I2V dimension. Dynamic degree improves in this single run, but semantic and image-quality losses dominate the aggregate.

only candidate would test whether the contiguous six-layer removal is already too destructive. A width-only candidate would test whether the 1344-hidden and 7168-FFN reductions dominate the observed image-quality losses. A less-aggressive combined candidate closer to the 1.0B ceiling would test whether the current 920.9M point is unnecessarily aggressive. A longer or stronger recovery schedule would test whether step 100 is too early to judge the architecture. Without those rows, the present paper can rank observed failures but cannot isolate the cause.

Claim boundary. The available evidence enforces three boundaries on the interpretation. First, the parameter claim is supported: the candidate is below 1.0B DiT parameters. Second, the quality-success claim is not supported: degradation is 18.944%, outside the target. Third, causal method claims are blocked: no ablation matrix exists. This separation is easy to lose in prose, because a paper can truthfully say that it built a smaller model while falsely implying that the smaller model is deployable. The manuscript keeps those claims separate.

The paired detail bootstrap diagnostics are useful for locating prompt-level and detail-level failure modes, but they are not multi-seed confidence intervals. They resample matched details inside the same completed run. The paper therefore reports them only as diagnostics and does not use them for aggregate significance language.

The bootstrap rows are still useful for engineering triage. They indicate that large negative deltas are concentrated in aesthetic quality, imaging quality, i2v subject, subject consistency, and i2v background. Those are the same dimensions that dominate the official per-dimension table. Agreement between the aggregate table and detail diagnostics increases confidence that the failure is not a formatting effect in one summary row, while still falling short of multi-run statistical evidence.

Next-candidate implication. The most direct next candidate is not a larger unbounded model, but a less aggressive sub-1B DiT point near the threshold. The current candidate is 920.9M parameters, leaving roughly 79.1M parameters before the 1.0B limit. That unused budget could be spent on retaining more width, retaining more feed forward capacity, restoring part of the middle layer interval, or choosing a different structured mapping. The evidence does not say which option is best. It does say that the current point spends less than the full parameter budget and loses too much quality, so the next experiment should use the remaining budget to test whether a gentler structured candidate improves the largest-loss dimensions.

Recovery implication. The evaluated checkpoint is a step-100 recovery pilot. A longer recovery run is not guaranteed to solve the loss, but the current evidence cannot rule it out. The correct claim is therefore conditional: this specific recovered checkpoint is a no-go; recovery as a strategy remains unresolved. A follow-up should pair architecture aggressiveness and recovery length, because changing only one axis may leave the project unable to distinguish architectural damage from under-recovery.

Failure Cases

The saved detail rows identify large losses in image aesthetics, image quality, subject faithfulness, and background faithfulness. Examples with the largest normalized losses include food, landscape, portrait, flower, and underwater prompts. These examples suggest that the current denoiser reduction damages fine visual detail and semantic conditioning more than it damages temporal smoothness. Because the paper has no qualitative video panel and no human preference study, the examples are treated as failure categories rather than claims about user perception.

The failure categories also clarify what kind of recovery signal may be needed. Losses in food, landscape, flower, portrait, and underwater examples point to textures, color, object boundaries, and image-conditioned semantics rather than only frame-to-frame continuity. A recovery objective that improves temporal smoothness alone would not address the measured weakness. Conversely, an architecture that preserves subject and imaging quality while slightly reducing dynamic degree might be preferable under the weighted aggregate, because the current candidate already demonstrates that high dynamic degree does not rescue poor semantic and visual scores.

A separate protocol failure mode remains unresolved. The official evaluator emitted missing-video warnings for expected suffix variants even though both generated video sets contain one video per prompt and zero missing generated videos. Until the one-video-per-prompt contract is justified for this VBench-I2V path, or the expected variants are regenerated and scored, the paper should present the score as official but protocol-warned evidence.

This warning affects the confidence level of the protocol, not the direction of the current manuscript claim. The manuscript's main claim is negative: the candidate does not satisfy the target under the completed official scoring run. If future protocol repair changes the absolute aggregate, the repaired score should replace the current score. Until then, the honest presentation is to report the official completed score and disclose the warning that could matter for final protocol certification.

Discussion

The result highlights a practical trap in structured pruning for generative video models: parameter-count success can arrive before benchmark-quality success. In discriminative compression, a single top-line accuracy drop often serves as the main accept/reject criterion. In image-to-video generation, the aggregate can hide divergent behavior across motion, temporal stability, semantic faithfulness, and image quality. The current candidate shows exactly that pattern. It nearly retains temporal flickering and motion smoothness and improves dynamic degree, yet loses badly on aesthetic, imaging, subject, and background dimensions.

For a deployment-oriented pruning paper, this means the acceptance criterion must be multi-dimensional from the start. A candidate that meets the parameter budget should not be advanced merely because one temporal metric is stable or one motion metric improves. It must also preserve the conditioning and visual quality dimensions that dominate user-visible I2V failures. The current evidence suggests that future candidates should be screened early on these semantic and visual dimensions before spending budget on full recovery and multi-seed scoring.

The no-go result also clarifies what would make a positive paper credible. A future success claim would need a candidate at or below 1.0B DiT parameters, the same non-DiT unchanged policy, official non-camera VBench-I2V degradation inside the 3–5% band, protocol resolution for expected video variants, and ablations that separate depth, width, recovery, and candidate aggressiveness. Without those pieces, a positive claim would be underdetermined even if one aggregate score looked acceptable.

Finally, the result is useful as a stopping rule. The evaluated point is far enough outside the quality band that additional prose polish cannot change its scientific status. The correct action is to preserve the no-go evidence, report the limitations, and define the next experiment around the measured bottleneck: semantic and visual-quality retention under the fixed DiT-only and non-DiT unchanged constraints.

Diagnostic Taxonomy

The evidence in this paper separates four failure types that should not be merged. The first is *target failure*: the candidate misses the pre-specified quality budget even though it satisfies the parameter budget. This is the only failure type directly proven by the aggregate score. The second is *dimension concentration*: losses are not evenly distributed, but instead concentrate in aesthetic quality, imaging quality, subject faithfulness, subject consistency, and background faithfulness. This is supported by the per-dimension table and the matched-detail diagnostics. The third is *causal ambiguity*: the current evidence cannot determine whether depth removal, width reduction, short recovery, or conditioning mismatch is responsible. The fourth is *protocol uncertainty*: the official evaluator completed and produced usable scores, but emitted missing-variant warnings that must be resolved before the protocol can be treated as final without qualification.

Keeping these categories separate improves both scientific honesty and engineering efficiency. If target failure were reported without dimension concentration, the next experiment would know only that the candidate is bad, not which metrics to repair. If dimension concentration were reported without causal ambiguity, the paper might overfit its explanation to the most visible losses and claim that a particular pruning axis caused them. If protocol uncertainty were ignored, a later reproduction could appear to contradict the paper for reasons unrelated to architecture. If protocol uncertainty were used to discard the completed score entirely, the paper would lose a real official measurement that currently rejects the candidate. The taxonomy therefore acts as a guardrail on interpretation.

The taxonomy also defines acceptance criteria for the next round. A stronger candidate must eliminate target failure by landing inside the 3–5% aggregate degradation band. It should reduce dimension concentration by improving the semantic and visual-quality rows that dominate the current loss. It should reduce causal ambiguity through at least one ablation axis, preferably comparing a less aggressive sub-1B candidate against the present point. It should reduce protocol uncertainty by either justifying the one-video-per-prompt I2V contract or regenerating the expected suffix variants. These criteria are derived from the measured bottlenecks in this run rather than from a generic compression wish list.

Conclusion

This paper documents a negative but actionable result for DiT-only structured pruning of Wan2.1-Fun-V1.1-1.3B-InP. The evaluated candidate keeps non-DiT modules unchanged and reaches 920,852,288 DiT parameters, below the 1.0B target. It fails the quality target: official non-camera VBench-I2V aggregate degradation is 18.944%, not 3–5%. The next paper-worthy step is therefore not polishing this candidate into a success story, but resolving the evaluator-variant warning and evaluating less-aggressive or longer-recovered structured candidates with the same matched protocol.

Limitations

The study has six binding limitations. First, it evaluates one candidate. The result does not rule out a different sub-1B DiT, especially one closer to the 1.0B boundary. Second, it uses one matched seed. The comparison is controlled within that seed, but it is not a multi-seed robustness claim. Third, the recovery checkpoint is step 100, and no longer-recovery evaluation is available. Fourth, the evidence does not include a depth-only or width-only ablation, so the paper cannot attribute the measured loss to one pruning axis. Fifth, the official evaluator emitted missing-variant warnings that must be resolved before the protocol can be treated as final without caveat. Sixth, no human preference study or qualitative video panel is included, so all quality statements are about the official benchmark dimensions reported here.

These limitations bound the implication of the no-go result. The result is strong enough to prevent a positive claim for the evaluated candidate. It is not strong enough to say that DiT-only structured pruning cannot work for Wan I2V. The next candidate should therefore be evaluated with the same non-DiT unchanged policy, the same camera-excluded official dimensions, and additional controls for aggressiveness, recovery length, and seeds.

Ethics Statement

This paper does not introduce a new dataset, collect human-subject data, or deploy a new generative system. The main ethical risk is misreporting model capability: presenting a smaller model as successful when the measured quality loss is outside the target would mislead downstream users. The manuscript mitigates that risk by reporting the no-go result directly, preserving the protocol warning, and avoiding claims of human perceptual equivalence.

Reproducibility Checklist

- The pruning surface is restricted to the DiT denoiser; non-DiT modules are recorded as unchanged references.
- The evaluated candidate has 920,852,288 measured DiT parameters.
- The official non-camera VBench-I2V evaluation contains 355 matched task rows and nine non-camera dimensions.
- Baseline and recovered/pruned conditions use matched prompts, images, generation settings, and seed 43.
- The paper does not claim target success because aggregate degradation is 18.944%.
- Current limitations are single-seed evidence, evaluator missing-variant warnings, no ablation matrix, no less-aggressive sub-1B candidate, and no longer-recovery evaluation.

The replay contract for the evaluated result consists of the public Wan model family, the official VBench-I2V non-camera dimensions, matched prompt/image/video conditions, seed 43, and the aggregate and per-dimension metrics reported above. Additional replay metadata can be provided as supplementary material without changing the scientific claims in the main text.

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Supplementary Reproducibility Details

The evaluated pruning surface is the DiT denoiser only. The non-DiT stack is kept fixed: VAE, text encoder, image encoder, tokenizer, scheduler, and other non-DiT components are treated as unpruned references to the same Wan I2V backend used by the baseline. The pruned candidate keeps 24 of 30 DiT layers, removes source layers 12–17, reduces hidden width from 1536 to 1344, reduces feed-forward width from 8960 to 7168, keeps 12 attention heads, and changes per-head width from 128 to 112.

The evaluated benchmark contract uses the official non-camera VBench-I2V dimensions listed in Table 2. Baseline and recovered candidate generations are matched by prompt, conditioning image, seed, frame count, denoising steps, resolution, and guidance. The appendix adds no new score beyond the main text; it records the replay assumptions needed to interpret the no-go result.